# CS 474 Image Processing and Interpretation

Jackson Loughmiller

Programming Assignment 1

# 1. Image Sampling

## Theory

Image sampling is a method generally used to convert an analog (continuous) image into a digital representation of that image. Since digital images cannot be represented continuously, it is necessary to select certain points from the analog image that will be represented as individual pixels in the digital image. When sampling an image that is already in digital form, the resulting image will be a smaller, less detailed version of the input.

In this part of the assignment, the sampling method used is sub-sampling, a technique that creates a new image by removing the rows and columns between every $n^{th}$ pixel, n being an integer specified by the user. For example, if n is 2, the output will be half the size of the input, and if n is 4, it will be one quarter of the size.

**Eq. 1.1** $(row - 1)mod(factor)$

**Eq. 1.2** $(col - 1)mod(factor)$

**Eq. 1.3** $output_row \ = row//factor$

**Eq. 1.4** $output_col \ = col//factor$

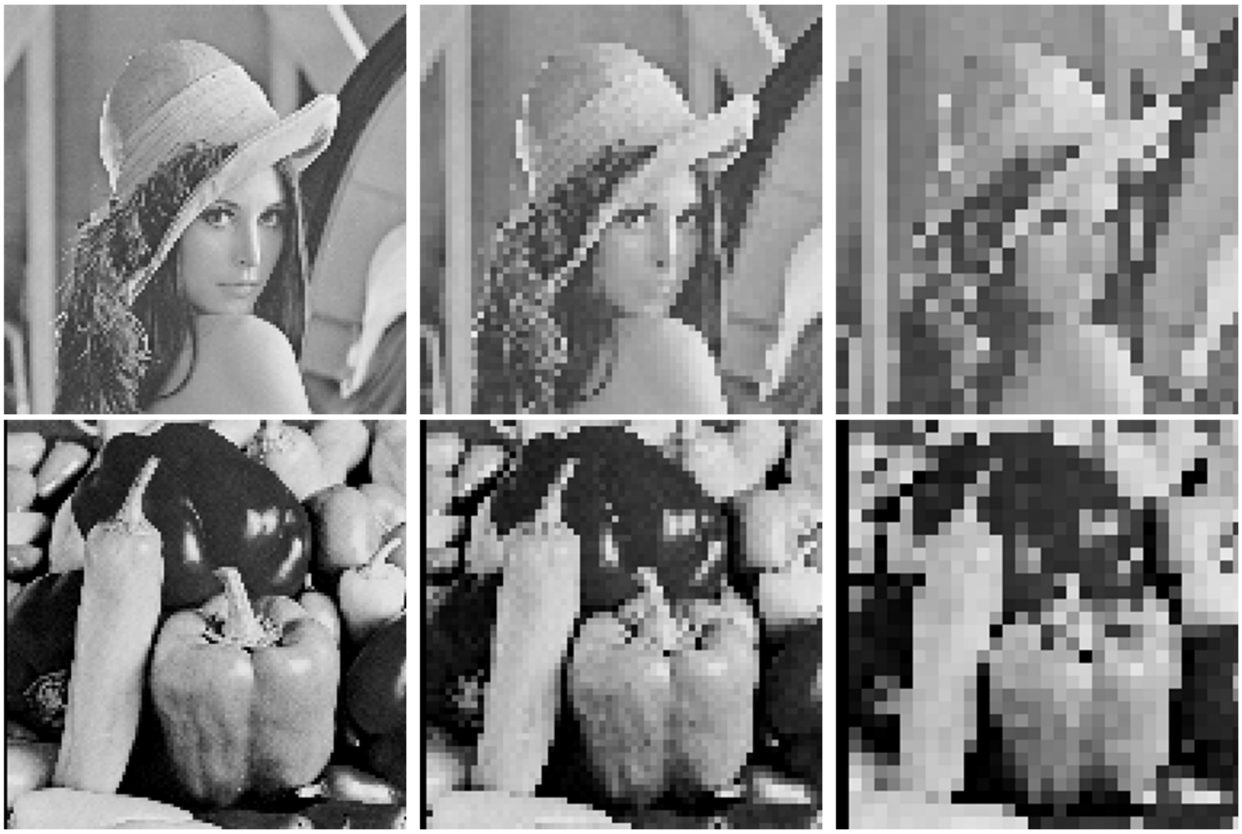
Equations 1.1 and 1.2 are used to decide whether a pixel from the input image is going to be sampled. If both equations result in 0, that pixel will be sampled and placed into the output image at the destination determined by equations 1.3 and 1.4.

## Results and Discussion

As seen below in Figure 1.2, as the factor for sub-sampling increased during testing, the image quality degraded. Once a factor of 8 was reached, it was extremely difficult to discern any small details from the image.

| A | B |
|---|---|

Figure 1.1: Original images to be used in sampling and quantization.



| | | |
|---|---|---|
| A | B | C |
| D | E | F |

Figure 1.2: Images subjected to different factors of sub-sampling, resized to 256x256. Sub-sampled factor of 2 (A, D), Sub-sampled factor of 4 (B, E), Sub-sampled factor of 8 (C, F)

# 2. Image Quantization

## Theory

Image quantization is a method to reduce the range of gray levels allowed in an image. In practice, quantization is used mainly to sample gray levels from an analog image to transform it into a digital image. In this project, the goal was simply to reduce the gray levels to see the effects of doing so. To reduce the levels, the following equations were used:

**Eq. 2.1** $$slope = \frac{gray_levels}{256}$$

Eq. 2.1 slope assumes an 8-bit image, with possible pixel values in the range [0, 255].

**Eq. 2.2** $$output_pixel = \frac{\lfloor slope \cdot input_pixel \rfloor}{slope}$$

The numerator of Eq. 2.2 represents the mapping of input values to the desired range of gray levels and is floored to make the output a valid pixel value. This value is then divided by the slope to map the values back to the full range of gray levels for better visualization of the results.

## Results and Discussion

As seen below in Figure 2.1, quantization has a less noticeable effect on image quality than sampling, in fact most details can still be made out even when the gray levels are reduced to 32, 8 times fewer possible pixel values than the original image. Artifacts start to show up more obviously when the gray levels are reduced to 8, and once reduced to 2 the image becomes binary. However, even the binary image is significantly more detailed than the same image subsampled at a factor of 8 (Figure 1.2 C and F).

| | | | |
|---|---|---|---|
| A | B | C | D |
| E | F | G | H |

Figure 2.1: Images subject to differing levels of quantization, remapped to pixel values ranging [0, 255] for visualization purposes. 128 gray levels (A, E), 32 gray levels (B, F), 8 gray levels (C, G), 2 gray levels (D, H).

# 3. Histogram Equalization

## Theory

Histogram equalization is a technique used to equalize the contrast of an image automatically, without the need to pass parameters other than the input image. The first step of histogram equalization is to calculate the PDF/CDF for the image, in this case the CDF since the input is an image with discrete pixel values. From the input image, the following are calculated:

**Eq. 3.1:** $$pixel_frequency = sum\ of\ all\ pixels\ of\ same\ value$$

**Eq. 3.2:** $$pixel_probability = \frac{pixel_frequency}{width \times height}$$

The histogram of the input image is generated by creating a bar chart with the possible pixel values on the x-axis, and the probability of a pixel being a certain value (calculated using Eq. 3.2) on the y-axis. The output values associated with each input pixel are then calculated using the following equation (Eq. 3.3), where *pixel_location* refers to the index of a certain pixel when the image is converted to a 1-dimensional array of values, and *denorm_cum_cdf* refers to the denormalized cumulative distribution function:

**Eq. 3.3:** $$denorm_cum_cdf = round\left((gray_levels - 1)\sum_{i=0}^{pixel_location} pixel_probability_i\right)$$

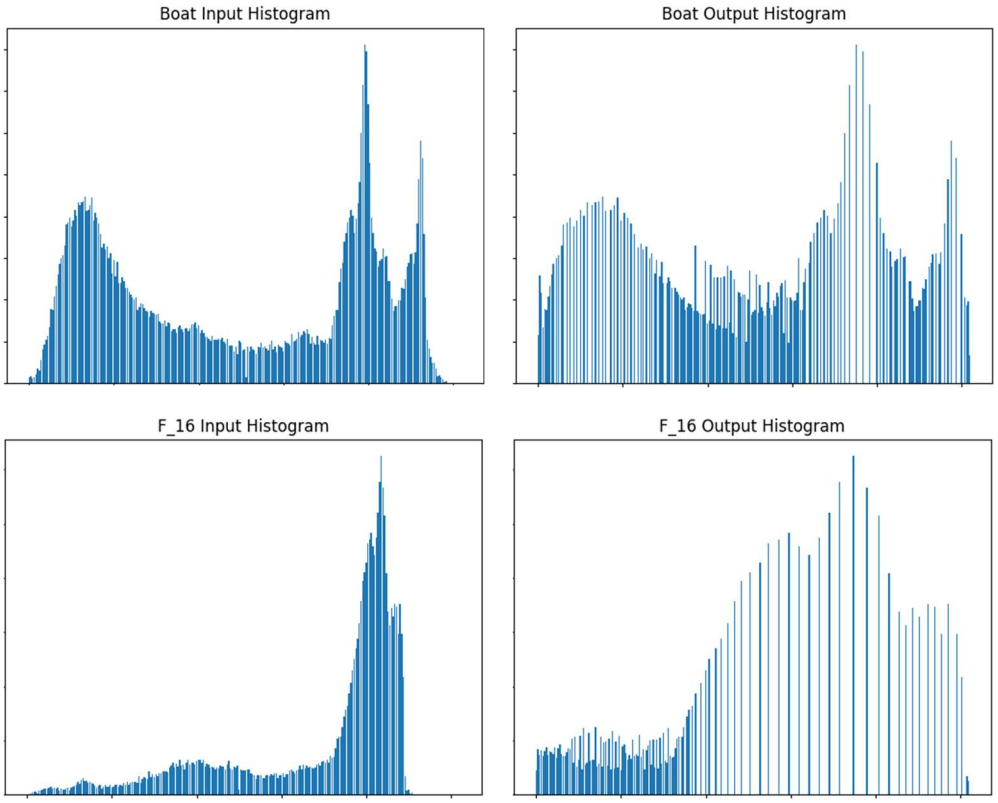
The value given by Eq. 3.3 is calculated for each input pixel. The equalized output image is then generated by mapping each value from the resulting array to a new image. The output histogram is generated using the same method as before with Eq.'s 3.1 and 3.2.

## Results and discussion

As seen below in Figure 3.1, histogram equalization can have differing effects depending on the contrast in the input image. The same process was applied to both of the following images, but there were more drastic contrast changes to the image shown in C and D, than the image in A and B. This is due to the first image already having a relatively equalized contrast to begin with, while the second image was concentrated more with high value pixels. This difference can be seen more clearly by looking at the input image histograms, Figure 3.2 A and C. The output histograms give a clear visual of the effects of equalization on each image. The histogram in Figure 3.2 B is not significantly more distributed than its input histogram in Figure 3.2 A. Conversely, the histogram in Figure 3.2 D is significantly more distributed it's input in 3.2 C. These differing histograms give a visual explanation of why the contrast changes are so vastly different in Figure 3.1 A to B, than from Figure 3.1 C to D.

| | |
|---|---|
| A | B |
| C | D |

Figure 3.1: Original images (A, C), Histogram Equalized Images (B, D)




| | |
|---|---|
| A | B |
| C | D |

Figure 3.2: Image histograms

X-axis: Pixel values

Y-axis: Probability